
This guideline is intended for use as a clinical reference and knowledge tool, and should be used in conjunction with individualised clinical judgement, local protocols, or specialist consultation by qualified healthcare professionals in the context of the patient's full clinical picture.

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1 Background & Global Context

Definition of Infertility

Infertility is defined as the failure to achieve a clinical pregnancy after 12 months or more of regular, unprotected sexual intercourse. For women aged 35 and above, this threshold is reduced to 6 months, given the accelerated age-related decline in ovarian reserve. Infertility may be classified as **primary** (no previous pregnancy has been achieved) or **secondary** (infertility occurring after at least one prior pregnancy, regardless of outcome).

Global and Regional Epidemiology

Infertility is a significant global public health concern affecting approximately **186 million people worldwide**, with an estimated 48 million couples experiencing difficulty conceiving at any given time (WHO, 2020). The condition does not discriminate by geography or socioeconomic status, but its impact and management options vary dramatically across regions.

Region / Statistic	Prevalence / Data	Source
Global couples affected	~48 million (at any time)	WHO 2020
Sub-Saharan Africa — primary infertility	3%–7% of women of reproductive age	Dhont et al., 2011
Sub-Saharan Africa — secondary infertility	Up to 30% in some communities	Larsen, 2000; AJOL
Tanzania — estimated prevalence	~8%–12% of reproductive-age women	MUHAS, 2018
Female factor contribution	~40% of all infertility cases	FIGO, 2019
Male factor contribution	~30%–40% of all cases	FIGO, 2019
Combined / unexplained	~20%–30%	ACOG, 2020

The Female Reproductive Cycle

A normal menstrual cycle lasts 21–35 days (average 28 days), comprising the follicular phase, ovulation, and the luteal phase. Ovulation — the release of a mature oocyte — typically occurs around day 14 in a 28-day cycle. Fertilisation must occur within 12–24 hours of ovulation. A woman is born with approximately 1–2 million oocytes; by puberty this declines to ~300,000–400,000, and only 300–500 will mature over a lifetime. Ovarian reserve declines progressively, with a marked acceleration after age 35 and again after 37.

Key physiological hormones involved in female fertility

- FSH (Follicle-Stimulating Hormone) — stimulates follicular growth in the ovary
- LH (Luteinising Hormone) — triggers ovulation; surges 24–36 hours before egg release
- Oestradiol (E2) — produced by developing follicles; prepares the uterine lining
- Progesterone — secreted by the corpus luteum after ovulation; maintains pregnancy
- AMH (Anti-Müllerian Hormone) — key marker of ovarian reserve; does not vary with cycle
- Prolactin — high levels suppress ovulation (hyperprolactinaemia causes infertility)
- TSH (Thyroid-Stimulating Hormone) — thyroid dysfunction disrupts the entire reproductive axis

2 Challenges Facing Women with Fertility Problems

2.1 Medical & Physiological Challenges

The physiological causes of female infertility span multiple organ systems and hormonal axes. Each presents its own diagnostic and therapeutic challenge:

Ovulatory dysfunction

The most common cause of female infertility (~25–30%). Includes PCOS, hypothalamic amenorrhoea, hyperprolactinaemia, premature ovarian insufficiency (POI), and thyroid disorders. Women may have absent, irregular, or unpredictable periods making timing of intercourse difficult.

Tubal factor infertility

Accounts for ~25–35% of female infertility in Sub-Saharan Africa — proportionally higher than in high-income settings. Damage or blockage of the fallopian tubes prevents egg and sperm from meeting. Primary causes: pelvic inflammatory disease (PID) from STIs, post-abortion sepsis, endometriosis, and prior pelvic surgery.

Uterine and endometrial factors

Structural abnormalities of the uterus impair implantation or increase miscarriage risk. Includes fibroids (especially submucosal), endometrial polyps, Asherman syndrome (intrauterine adhesions from D&C;), uterine septum, and adenomyosis.

Endometriosis

Affects ~10% of women of reproductive age globally; ~30–50% of women with endometriosis experience infertility. Ectopic endometrial tissue causes inflammation, adhesions, distorted pelvic anatomy, and impaired oocyte quality. Often significantly delayed in diagnosis (average 7–10 years).

Diminished ovarian reserve (DOR)

Natural age-related decline accelerated by genetics, autoimmune disease, prior ovarian surgery, or chemotherapy. Characterised by low AMH, high FSH, low antral follicle count (AFC). Significantly reduces response to ovarian stimulation.

Cervical factors

Hostile cervical mucus, cervical stenosis, or anti-sperm antibodies may prevent sperm from entering the uterus. Less common but treatable cause of infertility.

Unexplained infertility

~15–25% of couples complete a full investigation with no identified cause. Likely reflects limitations in diagnostic tools rather than absence of pathology. Empirical treatment is recommended.

2.2 Psychosocial & Cultural Challenges

Beyond the medical dimension, women face profound psychosocial burdens that are often underestimated by healthcare systems:

- Social stigma and community pressure — infertility carries significant social stigma, particularly in cultures where womanhood is strongly tied to motherhood. Women are frequently blamed regardless of the actual cause.
- Mental health impact — infertility is associated with elevated rates of depression (2–3× higher than fertile women), anxiety, and reduced quality of life. Grief responses include denial, anger, and social withdrawal.
- Marital stress and relationship breakdown — in many contexts, infertility leads to polygamy, divorce, or domestic conflict. Women bear disproportionate emotional and social consequences.
- Economic barriers to treatment — assisted reproductive technologies (ART) such as IVF remain unaffordable for most women in low- and middle-income countries. Out-of-pocket healthcare costs drive families into debt.
- Healthcare access barriers — long distances to specialist centres, limited trained reproductive health professionals, poor diagnostics infrastructure, and long waiting lists create delays in care.
- Cultural and religious beliefs — spiritual attribution of infertility may delay engagement with biomedical services. Women may pursue traditional remedies for years before seeking clinical care.
- Partner non-participation — male partners frequently refuse semen analysis, viewing it as threatening to masculinity. This delays diagnosis of male factor and places all diagnostic burden on the woman.
- Information gaps — low health literacy about the menstrual cycle, fertile window, and early signs of reproductive disorders leads to delayed presentation and self-blame.

WHO recognition of infertility as a public health priority

- WHO classifies infertility as a disease of the reproductive system (ICD-10: N97) requiring medical treatment
- The right to found a family is recognised in international human rights frameworks
- WHO Traditional Medicine Strategy 2014–2023 calls for integration of safe traditional practices
- Sustainable Development Goal 3.7 includes universal access to reproductive healthcare services

3 Signs & Symptoms

Infertility itself has only one definitive symptom: failure to conceive after the defined period. However, the underlying conditions causing infertility often present with recognisable signs and symptoms. Early recognition of these features enables timely investigation and intervention.

Polycystic Ovary Syndrome (PCOS)

- Irregular or absent periods (oligomenorrhoea / amenorrhoea) — cycles >35 days or <8 cycles/year
- Excessive hair growth on face, chest, abdomen, or back (hirsutism)
- Acne — persistent, often jawline or chin distribution
- Hair thinning or male-pattern baldness (androgenic alopecia)
- Unexplained weight gain, especially around the abdomen
- Skin darkening in skin folds (acanthosis nigricans) — sign of insulin resistance
- Multiple small follicles on ovaries seen on ultrasound ('string of pearls' sign)
- Elevated androgens on blood test (testosterone, DHEAS)
- Difficulty conceiving — anovulatory cycles

Endometriosis

- Severe, cyclical pelvic pain — worsens during menstruation (dysmenorrhoea)
- Deep dyspareunia — pain during or after intercourse, especially in certain positions
- Cyclical rectal pain or painful defecation during menstruation
- Cyclical haematuria (blood in urine) if bladder is affected
- Heavy menstrual bleeding (menorrhagia) or irregular spotting
- Fatigue — often disproportionate and cyclical
- Infertility — may present with no other symptoms in some women
- Bloating and gastrointestinal symptoms premenstrually

Pelvic Inflammatory Disease (PID) & Tubal Factor

- Lower abdominal or pelvic pain — may be bilateral, dull or cramping
- Abnormal vaginal discharge — purulent, malodorous
- Fever and chills (in acute PID)
- Pain during intercourse (dyspareunia)
- Irregular bleeding between periods
- Painful or frequent urination
- Nausea and vomiting in severe cases

■ **NOTE: Tubal damage from past PID may be entirely asymptomatic — infertility may be the only presenting feature**

Uterine Fibroids

- Heavy menstrual bleeding (menorrhagia) — may lead to anaemia
- Prolonged periods (>7 days)
- Pelvic pressure or heaviness — 'bloating' sensation
- Frequent urination if fibroid compresses bladder
- Constipation if fibroid compresses rectum
- Lower back pain
- Pain during intercourse
- Recurrent miscarriage or difficulty conceiving (submucosal fibroids particularly)
- Enlarged abdomen or palpable abdominal mass in large fibroids

Premature Ovarian Insufficiency (POI) / Diminished Ovarian Reserve

- Irregular or absent periods before age 40
- Hot flashes and night sweats (menopausal symptoms in a young woman)
- Vaginal dryness and dyspareunia
- Mood changes, irritability, difficulty concentrating
- Reduced libido
- Elevated FSH (>25 IU/L on 2 occasions, 4 weeks apart)
- Very low AMH (<0.5 ng/mL)
- Fatigue and sleep disturbance

Thyroid Disorders Affecting Fertility

- Hypothyroidism: weight gain, fatigue, cold intolerance, constipation, dry skin, hair loss, heavy periods, depression
- Hyperthyroidism: weight loss, heat intolerance, palpitations, anxiety, diarrhoea, light/absent periods
- Both conditions cause ovulatory dysfunction and increase miscarriage risk
- Anti-thyroid antibodies (especially anti-TPO) associated with recurrent miscarriage even with normal TSH

Hyperprolactinaemia

- Galactorrhoea — milky nipple discharge outside pregnancy/breastfeeding
- Irregular or absent periods
- Reduced libido
- Headaches or visual field changes (if caused by pituitary adenoma)
- Infertility from suppressed ovulation

4 Diagnosis — Clinical Assessment Pathway

4.1 History Taking

A thorough history is the foundation of infertility assessment:

- Duration of infertility and frequency/timing of intercourse
- Menstrual history: cycle length, regularity, flow, dysmenorrhoea, intermenstrual bleeding
- Obstetric history: previous pregnancies, miscarriages, terminations, ectopic pregnancies
- Past medical history: STIs, PID, pelvic/abdominal surgery, autoimmune conditions, thyroid disease
- Medications: current medications, prior hormonal contraception, chemotherapy, steroids
- Family history: POI, PCOS, chromosomal abnormalities, early menopause
- Social history: smoking, alcohol, drug use, BMI, occupation, exercise habits, stress
- Sexual history: dyspareunia, libido, contraception history

4.2 Physical Examination

- BMI and weight distribution (central obesity in PCOS)
- Signs of hyperandrogenism: hirsutism (Ferriman-Gallwey score), acne, alopecia
- Signs of thyroid dysfunction: goitre, tachycardia, skin/hair changes
- Breast examination: galactorrhoea (check for discharge)
- Abdominal examination: masses, tenderness, organomegaly
- Pelvic examination: uterine size and regularity, adnexal tenderness or masses, cervical motion tenderness (sign of PID)
- Acanthosis nigricans (insulin resistance in PCOS)

4.3 Investigations — Recommended Panel

Investigation	Timing	What it assesses	Normal / Key values
FSH, LH	Day 2–3 of cycle	Ovarian reserve, pituitary function	FSH <10 IU/L; LH/FSH ratio >2 in PCOS
Oestradiol (E2)	Day 2–3	Baseline ovarian activity	<80 pmol/L (normal baseline)
AMH	Any day in cycle	Ovarian reserve (best single marker)	>1.0 ng/mL adequate; <0.5 ng/mL poor reserve
Prolactin	Morning, fasting	Hyperprolactinaemia screening	<500 mIU/L normal
TSH, fT4	Any time	Thyroid function	TSH 0.4–4.0 mIU/L
Testosterone (total)	Morning	Androgen excess (PCOS, adrenal)	<2.0 nmol/L
Progesterone (mid-luteal)	Day 21 (28-day cycle)	Confirms ovulation	>16–25 nmol/L confirms ovulation

FBC, HbA1c, lipids	Any time	General health, insulin resistance	HbA1c <5.7% normal
Pelvic ultrasound	Day 2–5 (follicular)	AFC, uterine morphology, ovarian cysts	AFC >10 per ovary = good reserve
Hysterosalpingography (HSG)	Day 7–10 post-period	Tubal patency, uterine cavity	Bilateral fill and spill = patent tubes
Semen analysis (partner)	2–5 days abstinence	Male factor — must be included	WHO 2021 reference values
Diagnostic laparoscopy	If indicated	Endometriosis, adhesions, tubal assessment	Gold standard for endometriosis

5 Treatment Procedures

5.1 Lifestyle & Non-Pharmacological Interventions (First-Line)

Before pharmacological or surgical treatment, modifiable lifestyle factors should be addressed. These interventions are low-cost, safe, and evidence-supported:

→ Weight optimisation

Even 5–10% weight reduction in overweight women with PCOS restores spontaneous ovulation in 55–65% of cases. BMI >30 significantly reduces response to fertility treatment. Weight gain in underweight women with hypothalamic amenorrhoea can fully restore ovulation.

→ Smoking cessation

Smoking accelerates follicle depletion, reduces IVF success rates by ~30%, and increases miscarriage risk. Cessation is mandatory before fertility treatment.

→ Alcohol reduction

Alcohol consumption >4 units/week is associated with reduced fecundity and increased miscarriage risk. Abstinence recommended when actively trying to conceive.

→ Nutritional optimisation

Mediterranean-style diet rich in antioxidants, lean protein, whole grains, and omega-3 fatty acids is associated with better fertility outcomes. Folic acid 400–800 mcg/day pre-conception is mandatory (reduces neural tube defects).

→ Stress management

Chronic psychological stress activates the HPA axis, suppressing GnRH and disrupting the HPG axis. Mindfulness, cognitive behavioural therapy (CBT), and counselling all show benefit in improving pregnancy rates in women undergoing ART.

→ Exercise

Moderate exercise is beneficial. However, high-intensity training (>60 min/day) in normal-weight women may suppress ovulation via hypothalamic disruption. Women with PCOS benefit significantly from structured aerobic exercise.

→ Optimise timing of intercourse

Intercourse every 2–3 days throughout the cycle is as effective as timed intercourse. LH-surge monitoring kits or basal body temperature charting can help identify the fertile window (5 days before and including ovulation day).

5.2 Ovulation Induction (OI)

Ovulation induction is indicated for women who are not ovulating (anovulation) or ovulating infrequently (oligoovulation):

Condition	First-line treatment	Notes
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PCOS (WHO Group II)	Letrozole 2.5–5 mg (days 2–6) or Clomiphene citrate 50–100 mg (days 2–6)	Letrozole now preferred over clomiphene — higher live birth rate, lower multiple pregnancy rate
Hypothalamic amenorrhoea (WHO Group I)	Gonadotrophins (FSH/LH) with monitoring	Address underlying cause first (weight, stress). Pulsatile GnRH where available
Hypogonadotrophic hypogonadism	Gonadotrophins (FSH injections + LH/hCG)	Requires careful monitoring to prevent OHSS
Clomiphene-resistant PCOS	Letrozole, gonadotrophins, or laparoscopic ovarian drilling (LOD)	LOD reduces androgen levels and restores ovulation in ~50% without multiple pregnancy risk

5.3 Surgical Procedures

→ Laparoscopic ovarian drilling (LOD)

For clomiphene-resistant PCOS. Electrosurgical or laser puncture of ovarian follicles reduces androgen production and restores ovulation in ~50% of women. Avoids multiple pregnancy risk of gonadotrophins.

→ Tubal surgery

Salpingostomy or fimbrioplasty for distal tubal blockage. Success depends on extent of damage. Proximal blockage may respond to hysteroscopic tubal cannulation. Severely damaged tubes (hydrosalpinx) should be removed before IVF — they reduce IVF success by 50%.

→ Myomectomy

Surgical removal of fibroids. Hysteroscopic approach for submucosal fibroids; laparoscopic or open for intramural/subserosal. Recommended when fibroids are likely impairing fertility or implantation.

→ Hysteroscopic surgery

Treatment of intrauterine adhesions (Asherman syndrome), endometrial polyps, uterine septum. Improves uterine cavity receptivity for implantation.

→ Endometriosis surgery

Laparoscopic excision of endometriotic lesions, cysts (endometriomas), and adhesions improves spontaneous conception rates and IVF outcomes. Staged approach: surgery first, then fertility treatment if needed.

→ Cervical dilatation

For cervical stenosis causing haematometra or sperm access problems.

5.4 Assisted Reproductive Technologies (ART)

→ Intrauterine Insemination (IUI)

Processed sperm placed directly into the uterine cavity at the time of ovulation. Used for unexplained infertility, mild male factor, donor sperm, or cervical factor. Success rate per cycle: 10–20%. Usually offered for up to 3–6 cycles before escalating to IVF.

→ **In Vitro Fertilisation (IVF)**

Ovarian stimulation with gonadotrophins → egg retrieval (transvaginal ultrasound-guided) → laboratory fertilisation → embryo transfer. Success rates: ~35–40% per cycle in women <35; ~10–15% in women >42. Indicated for tubal factor, severe endometriosis, failed OI/IUI, male factor, unexplained infertility after 2–3 years.

→ **Intracytoplasmic Sperm Injection (ICSI)**

Single sperm injected directly into the egg. Used for severe male factor infertility (oligospermia, azoospermia via surgical sperm retrieval), failed fertilisation in prior IVF, or poor egg quality.

→ **Donor oocyte IVF**

Indicated for POI, very poor ovarian reserve, repeated IVF failure, genetic conditions. High success rates (50–65%) as donor eggs are typically from young, healthy women.

→ **Embryo cryopreservation**

Spare embryos frozen for future use. Allows segmented IVF cycles — stimulation/retrieval in one cycle, frozen embryo transfer in a subsequent natural or prepared cycle. Reduces OHSS risk.

→ **Preimplantation Genetic Testing (PGT)**

Biopsy of day-5 blastocyst to screen for chromosomal abnormalities (PGT-A) or specific genetic disorders (PGT-M) before transfer. Indicated for recurrent miscarriage, advanced maternal age, or known genetic conditions.

6

Clinical Medicines & Pharmacological Interventions

All medications below should be prescribed by a qualified clinician. Doses shown are standard reference doses — individual titration based on response and monitoring is required.

Drug	Class / Mechanism	Indication	Dose	Key cautions
Clomiphene citrate	Selective oestrogen receptor modulator (SERM); blocks negative feedback → increases FSH	Anovulatory infertility, PCOS (second-line after letrozole)	50 mg days 2–6 of cycle; increase by 50 mg each cycle if no response; max 150 mg	Thin endometrium, visual symptoms (stop if occur), twin rate ~8%, max 6 cycles
Letrozole	Aromatase inhibitor; reduces oestrogen → increases FSH; no endometrial thinning	PCOS — first-line ovulation induction; clomiphene-resistant anovulation	2.5 mg days 2–6; increase to 5 mg or 7.5 mg if no ovulation	Preferred in PCOS: higher live birth rate, lower multiple rate than clomiphene
Follicle-Stimulating Hormone (FSH) injections (Gonal-F, Puregon)	Exogenous gonadotrophin; directly stimulates follicular development	WHO Group I & II anovulation, IUI/IVF stimulation, clomiphene-resistant PCOS	37.5–75 IU/day SC; titrate based on ultrasound monitoring; trigger with hCG or GnRH agonist	OHSS risk — requires careful monitoring; multiple pregnancy risk
hCG (human chorionic gonadotrophin) (Ovitrelle, Pregnyl)	LH analogue; triggers final oocyte maturation and ovulation	Ovulation trigger in OI and IVF cycles when follicle(s) reach 17–20 mm	5,000–10,000 IU IM or 250 mcg SC (recombinant); single dose	Contra in OHSS risk; withhold if >3 follicles >14mm in OI cycles
Metformin	Biguanide; insulin sensitiser; reduces hyperinsulinaemia and androgen levels in PCOS	PCOS with insulin resistance; adjunct to letrozole/clomiphene; reduces OHSS risk in IVF	500 mg BD with food → increase to 1,000–1,500 mg/day over 4 weeks	GI side effects; lactic acidosis (rare); stop before IV contrast; continue in pregnancy if started pre-conception
Cabergoline	Dopamine agonist D2; suppresses prolactin secretion from pituitary	Hyperprolactinaemia — restores ovulation; also used to prevent/treat OHSS	0.25–0.5 mg twice weekly; increase based on prolactin levels	First-line over bromocriptine (better tolerated, once-weekly dosing). Headache, nausea

Levothyroxine (L-T4)	Synthetic thyroid hormone; replaces deficient T4	Hypothyroidism (overt or subclinical) in women trying to conceive (TSH target <2.5 mIU/L pre-conception)	50–100 mcg daily; titrate to TSH <2.5 mIU/L	Dose often increases in pregnancy (by 25–30%); monitor 4-weekly in T1
Progesterone (vaginal / IM)	Natural progestogen; maintains endometrium for implantation; luteal phase support	Luteal phase insufficiency; IVF/IUI luteal support; recurrent miscarriage with proven deficiency	Cyclogest 400 mg vaginally BD (IVF); or Utrogestan 200 mg vaginally TID	Continue until 12 weeks gestation in IVF. Do not use oral micronised progesterone for luteal support in natural cycles as evidence is weak
GnRH agonist (Buserelin, Leuprorelin)	Initially stimulates then downregulates pituitary; prevents premature LH surge in IVF	IVF protocols (long protocol downregulation); endometriosis symptom management	Long protocol: start day 21 of preceding cycle, 200 mcg intranasal TID or 0.5 mg SC daily	Menopausal side effects (hot flushes, bone density loss) with long-term use. Bone protection needed >6 months
GnRH antagonist (Cetrorelix, Ganirelix)	Immediate pituitary suppression without flare; prevents LH surge in stimulated cycles	IVF antagonist protocol — more flexible, shorter, lower OHSS risk than agonist protocol	0.25 mg SC daily from day 5–6 of stimulation (when lead follicle 12–14 mm) until trigger	Preferred protocol in women at OHSS risk (PCOS). Injection site reactions
PID Regimen: Ceftriaxone + Doxycycline + Metronidazole	Cephalosporin + tetracycline + nitroimidazole combination covers gonorrhoea, chlamydia, and anaerobes	Pelvic inflammatory disease — to prevent tubal scarring and preserve fertility	Ceftriaxone 500 mg IM stat + Doxycycline 100 mg PO BD x14 days + Metronidazole 400 mg PO BD x14 days	WHO 2016 STI treatment guidelines. Treat partner simultaneously. Test of cure at 72 hours
Folic acid	B vitamin; essential for neural tube development and DNA synthesis	All women planning pregnancy; mandatory pre-conception supplementation	400 mcg/day standard; 5 mg/day for women on anti-epileptics, prior NTD, or BMI >30	Start at least 1 month before conception and continue through T1. Safe, essential
Vitamin D	Fat-soluble hormone; receptors on ovary, endometrium, and pituitary	Deficiency associated with PCOS, reduced IVF success, recurrent miscarriage	1,000–2,000 IU/day (maintain serum 25-OH-D >75 nmol/L for fertility)	Check baseline 25-OH-D. Loading dose may be needed if severely deficient

CoQ10 (Ubiquinol)	Mitochondrial antioxidant; supports oocyte energy metabolism and quality	Poor ovarian reserve, advanced maternal age, to improve egg quality before IVF	200–600 mg/day ubiquinol form; begin 3–6 months before planned retrieval	Generally safe. Evidence base growing but not yet Grade A. No significant drug interactions
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7 Things to Avoid — Risk Factors & Prevention

Many causes of infertility are preventable. The following are evidence-based risk factors that women and couples should actively avoid or manage to protect reproductive health:

Lifestyle Factors That Impair Fertility

- ✗ Smoking — damages oocyte DNA, depletes ovarian reserve prematurely, reduces IVF success by up to 30%, doubles miscarriage risk. There is no safe level of smoking for fertility.
- ✗ Excessive alcohol — >4 units/week associated with reduced fecundity. Alcohol during early pregnancy risks fetal alcohol spectrum disorder. Abstinence recommended when trying to conceive.
- ✗ Recreational drugs — cannabis disrupts the HPG axis and impairs implantation. Cocaine and opioids suppress gonadotrophins. Anabolic steroids suppress sperm production in partners.
- ✗ Extreme exercise — high-intensity training in normal-weight women suppresses GnRH pulsatility, causing hypothalamic amenorrhoea. Balance exercise with adequate caloric intake.
- ✗ Very low body weight (BMI <18.5) — causes hypothalamic suppression, anovulation, and oestrogen deficiency. Associated with eating disorders (anorexia nervosa has highest infertility rate of any psychiatric condition).
- ✗ Severe obesity (BMI >35) — causes insulin resistance, elevated androgens, anovulation in PCOS, reduces IVF response, and increases miscarriage and obstetric complication risk.
- ✗ Chronic stress — sustained cortisol elevation suppresses GnRH and disrupts the HPG axis. Psychological support is a clinical intervention, not an optional extra.
- ✗ Excessive caffeine — >300 mg/day (approximately 3 espresso shots) associated with reduced fecundity and increased miscarriage risk. Reduce to <200 mg/day when trying to conceive.

Infectious & Environmental Risks

- ✗ Unprotected sexual intercourse with multiple partners — primary risk factor for STIs (chlamydia, gonorrhoea, Trichomonas) that cause PID and tubal infertility. Consistent condom use is protective.
- ✗ Delayed treatment of STIs — Chlamydia trachomatis and Neisseria gonorrhoeae are often asymptomatic in women. Annual STI screening recommended for sexually active women <25 and higher-risk groups.
- ✗ Post-abortion sepsis — unsafe abortion is a leading cause of secondary infertility in low-resource settings. Access to safe abortion services directly protects future fertility.
- ✗ Endocrine-disrupting chemicals (EDCs) — bisphenol A (BPA) in plastics, phthalates in cosmetics/food packaging, pesticides (organochlorines), and heavy metals impair hormone function. Reduce exposure where possible.
- ✗ Occupational hazards — excessive heat (for male partner — impairs sperm), radiation, heavy metal exposure (lead, mercury), and certain solvents impair reproductive function.
- ✗ Poorly controlled chronic infections — HIV, malaria, and tuberculosis all impair reproductive function through chronic inflammation, malnutrition, and direct hormonal disruption.

Medical & Iatrogenic Risks

- ✗ Delayed fertility treatment-seeking — women over 35 lose ~1–2% of remaining oocytes per month. Every 6-month delay above 38 significantly reduces ART success probability.
- ✗ Unnecessary pelvic surgery — any pelvic or ovarian surgery risks adhesion formation and reduced ovarian reserve. Surgeons should use anti-adhesion barriers and minimise ovarian tissue removal.
- ✗ Aggressive ovarian cystectomy — removing ovarian endometriomas >4 cm reduces ovarian reserve. Freeze eggs before surgery if reserve is already reduced; consider drainage rather than excision.
- ✗ Uncontrolled chronic diseases — poorly managed diabetes, autoimmune disorders (lupus, antiphospholipid syndrome), thyroid disease, and inflammatory bowel disease all impair fertility.
- ✗ Certain medications — chemotherapy (especially alkylating agents) causes premature ovarian failure. Fertility preservation (egg/embryo freezing) should be offered before cancer treatment. NSAIDs taken around ovulation may delay or prevent egg release (luteinised unruptured follicle syndrome — avoid days 10–17 of cycle).
- ✗ Long-term hormonal contraception concerns — while fertility returns after stopping all modern contraceptives, very long-term use of depot medroxyprogesterone acetate (Depo-Provera) may delay return of fertility by 6–12 months compared to other methods.
- ✗ Self-medication with traditional abortifacients — unripe papaya, high-dose Aloe latex, Carica papaya seeds, and various herbal concoctions used to induce or prevent pregnancy can cause uterine damage, sepsis, and secondary infertility.
- ✗ Ignoring recurrent miscarriage — two or more consecutive miscarriages warrant investigation (antiphospholipid syndrome, thrombophilias, uterine anomalies, chromosomal translocations in either partner). Early investigation significantly improves outcomes.

Dietary & Nutritional Avoidance

- ✗ Very low-fat diets — oestrogen is synthesised from cholesterol. Severe fat restriction impairs sex steroid production and disrupts menstrual cycles.
- ✗ High-glycaemic index (GI) diets — promote hyperinsulinaemia, exacerbate PCOS, and increase androgen levels. Replace refined carbohydrates with whole grains, legumes, and vegetables.
- ✗ Trans fats (partially hydrogenated oils) — associated with anovulatory infertility in prospective studies. Found in processed foods, fast food, and commercially baked goods.
- ✗ High processed meat consumption — associated with reduced fertility. Replace with fish (especially oily fish for omega-3), poultry, and plant proteins.
- ✗ Soy in very high quantities — contains phytoestrogens. Normal dietary consumption safe. Very high supplemental soy isoflavone doses may theoretically affect hormonal balance (evidence limited but caution in women with hormone-sensitive conditions).
- ✗ Micronutrient deficiencies — iron deficiency anaemia impairs oocyte quality. Zinc deficiency affects ovulation. Selenium and iodine deficiency impair thyroid function. Ensure diet covers all micronutrients or supplement appropriately.

8 When to Refer — Specialist Care Criteria

The following criteria indicate when a woman should be referred to a specialist reproductive medicine unit, gynaecologist, or fertility centre:

Refer immediately

- Suspected ectopic pregnancy — acute pelvic pain + positive pregnancy test
- Severe acute PID with peritonism, fever $>38.5^{\circ}\text{C}$, or failure to improve on oral antibiotics within 72 hours
- Suspected ovarian torsion — acute severe unilateral pelvic pain with adnexal mass on ultrasound
- Signs of premature ovarian insufficiency in a woman <40 — missed periods + menopausal symptoms

Refer after initial assessment

- No conception after 12 months of regular unprotected intercourse (6 months if age >35)
- Confirmed bilateral tubal occlusion on HSG
- Severe male factor infertility (azoospermia, severe oligozoospermia) on semen analysis
- Known or suspected endometriosis requiring laparoscopy
- Recurrent miscarriage (2+ pregnancy losses)
- Uterine fibroids likely affecting fertility or implantation
- AMH <0.5 ng/mL or FSH >10 IU/L in a woman <38 wishing to conceive
- Failure to ovulate after 3 cycles of clomiphene at maximum dose

Consider referral for specialist input

- PCOS not responding to 3–6 cycles of ovulation induction
- Hyperprolactinaemia on prolactin >1000 mIU/L or suspected pituitary adenoma
- Thyroid disorder with TSH >4.0 or <0.4 when trying to conceive
- BMI >35 with infertility — specialist weight management before ART
- Psychological distress significantly impacting quality of life or treatment engagement
- Desire to preserve fertility before cancer treatment or significant surgery

9 References & Evidence Base

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